Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a PPS-Based High-Performance Composite and Large-Scale 3D Printing Hybrid Chock Liner for Marine Applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Shipbuilding/marine system	*Main/auxiliary & engine system*	*Polymeric Materials*	*Composite Material Manufacturing Technologies*
Concept of Proposed Technology	The proposed technology aims to develop a **hybrid manufacturing technology that combines PPS-based composites with large-scale 3D printing** for ship chock liners used in the installation of marine engines and propulsion systems. Conventional chock liners are typically manufactured by machining PA-based blocks or by using epoxy-based grouting methods. However, these approaches suffer from inherent limitations, including surface temperature rise under high load and high friction conditions, degradation of dimensional stability due to wear, and difficulties in achieving precise installation accuracy. To address these issues, the proposed technology introduces a **novel chock liner structural concept**			

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	that integrates a **PPS UD CFRTP skin with high wear resistance and heat resistance** and a **PPS-based 3D-printed structure that enables customized geometry**. This hybrid structure is designed to overcome the limitations of conventional chock liners while improving durability and installation performance. The core technical concept of this project lies in applying a **PPS-based UD CFRTP skin to the contact surface** of the chock liner to ensure thermal and mechanical stability under high-friction operating conditions. At the same time, a **PPS structure fabricated by large-scale 3D printing** is applied to the lower section to achieve both geometry customization according to installation conditions and effective load distribution. In particular, the **joining technology between the composite skin and the 3D-printed structure** represents a key technological challenge of this project. Through international joint research, this project aims to establish a **new composite–3D printing joining concept** that can replace existing joining technologies currently dependent on overseas proprietary patents.
Rationale for Joint R&D Project	This international joint R&D project is driven by the need to overcome the technical limitations of conventional manufacturing methods for ship chock liners used in marine engine and propulsion system installations. Currently, chock liners are mainly produced using machined PA-based blocks or epoxy-based grouting methods. Under high-load and high-friction operating conditions, these approaches frequently experience surface temperature increases, wear-induced dimensional changes, and difficulties in maintaining installation accuracy, leading to performance degradation and increased maintenance costs. The **PPS-based composite–3D printing hybrid chock liner technology** proposed in this project offers a promising solution to fundamentally improve the wear resistance and heat resistance of existing chock liners. However, the **joining technology required to integrate composite skins with 3D-printed structures** represents a high-level technical challenge that is difficult to develop independently within a short period. In particular, achieving stable interfacial performance under high-temperature and high-load conditions and establishing reliable joining processes require extensive research experience and advanced technical expertise. Currently, composite–3D printing joining technologies are applied in a limited manner due to reliance on proprietary technologies patented by overseas companies, and the development of alternative technologies has become increasingly urgent. To address this challenge, the proposed project adopts a joint R&D framework that combines the **expertise of leading overseas research institutes in composite–additive manufacturing joining**, the **large-scale 3D printing-based design and manufacturing capabilities of domestic companies**, and the **PPS UD CFRTP forming and material characterization capabilities of domestic research institutes**. Through this collaborative approach, the project aims to internalize composite–3D printing joining-based chock liner manufacturing technologies and to validate prototype manufacturing and process applicability for marine industry applications.
Key Words	1. Ship Chock Liner 2. PPS-Based Composite 3. UD CFRTP Skin

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	4. Large-Scale 3D Printing 5. Robot-based Composite-3D Printing Joining Technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	The objective of this R&D project is to establish a **hybrid manufacturing technology that combines PPS-based composites with large-scale 3D printing** for ship chock liners used in marine engine and propulsion system installations, and to enhance the technology maturity to a level suitable for **practical application through prototype development**. To achieve this goal, the project will conduct comprehensive R&D activities covering the entire development chain, including design, materials, manufacturing processes, joining technologies, and prototype fabrication. 1. **To establish structural design technology for PPS-based composite–3D printing hybrid chock liners.** The basic structure of the chock liner will be defined based on marine engine installation conditions, and a structural design technology integrating a PPS UD CFRTP skin with a PPS 3D-printed structure will be established. 2. **To develop forming process technology for PPS UD CFRTP skins.** Lamination and forming processes for PPS UD CFRTP skins applied to the contact surface of the chock liner will be developed, and forming conditions that satisfy the wear resistance and heat resistance required under high-friction environments will be derived. 3. **To develop joining technology between composite skins and 3D-printed structures.** Joining processes and jig concepts enabling stable integration of PPS-based composite skins with 3D-printed structures will be developed, and the structural reliability of the joining interface will be secured. 4. **To establish manufacturing process technology for chock liners based on large-scale 3D printing.** Chock liner structures with geometry customization according to installation conditions will be fabricated using large-scale 3D printing, and a manufacturing process compatible with composite skin forming will be established. 5. **To fabricate chock liner prototypes and verify their applicability.** Prototype chock liners will be manufactured by integrating the developed design, material, and process technologies, and their applicability and manufacturability will be verified with consideration for real ship installation environments.			
R&D Target Specifications	• Compressive strength: ≥ 100 MPa (ASTM D695) • Flexural strength: ≥ 150 MPa (ASTM D790) • Heat deflection temperature (HDT): ≥ 200 °C (ASTM D648) • Thermal cycling stability: No deformation or interfacial delamination (IEC 60068-2-14) • Oil resistance: Strength reduction ≤ 10% (ISO 1817) • Saltwater resistance: No structural damage (ASTM B117)			
R&D Plan	**Phase 1 (Basic Research Phase, 6 months)** • **3D Factory (Korean SME Lead Organization)**			

- Analyze the application environment of ship chock liners and establish the basic design concept of a PPS-based composite–3D printing hybrid structure.
- Derive the basic geometry and structural design concepts for chock liners using large-scale 3D printing.

- **Fraunhofer ICT (International Partner)**
  - Define the joining concept between PPS-based composites and 3D-printed structures and conduct fundamental studies on the joining mechanism.
  - Identify candidate joining approaches considering PPS material characteristics and establish basic data and design guidelines for subsequent process development.
- **UNIST (Contracting Research Institute)**
  - Analyze the fundamental forming characteristics of PPS UD CFRTP skins and evaluate the applicability of IR heating and compression molding-based forming processes.
  - Assess the feasibility of achieving the basic material properties required for the contact surface.

Phase 2 - 1year (12months)

- **3D Factory (Korean SME Lead Organization)**
  - Review fixation structures and installation methods for chock liners, and develop geometry customization design technologies for 3D-printed structures.
  - Further refine the chock liner structural design with consideration of the joining process.
- **Fraunhofer ICT (International Partner)**
  - Analysis of PPS composite 3D printing interfacial characteristics based on the phase 1 preliminary research.
  - Elaborate the joining process concept between PPS-based composites and 3D-printed structures and evaluate the structural stability.
  - Conduct the goal and scope definition for the LCA screening on the joining process
- **UNIST (Contracting Research Institute)**
  - Conduct material property evaluation and forming condition analysis for PPS UD CFRTP skins.
  - Derive basic forming conditions for applying the skin to the contact surface

Phase 2 - 2year (12months)

- **3D Factory (Korean SME Lead Organization)**
  - Further develop the chock liner structure integrating the contact surface and the 3D-printed structure to a manufacturable level.
  - Establish a manufacturing process that integrates large-scale 3D printing with composite skin forming.
- **Fraunhofer ICT (International Partner)**
  - Develop the composite–3D printing joining process and verify their applicability to manufacturing processes. Serve as the advisory role for joining jig concept.
  - Implement technical improvements to ensure process reproducibility and stability.
  - Environmental impact assessment with LCA screening for joining process and goal and scope for the prototype ship chock liner
- **UNIST (Contracting Research Institute)**
  - Develop the forming process for PPS UD CFRTP skins applied to the contact surface and verify their structural applicability.
  - Secure process compatibility between the skin forming process and the joining process.

Phase 2 - 3year (6months)

- **3D Factory (Korean SME Lead Organization)**
  - Integrate the developed design, manufacturing, and joining technologies to fabricate the final chock liner prototype.
  - Verify product completeness and manufacturability with consideration for industrial application.
- **Fraunhofer ICT (International Partner)**
  - Conduct technology transfer in an advisory role and conduct process documentation for the developed composite–3D printing joining technologies.
  - Provide final technical documentation for joining system and serve as the advisory role for jig design.

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- Comparative screening LCA of the prototype ship chock liner conduction for environmental impact assessment.

- **UNIST (Contracting Research Institute)**
  - Optimize the PPS UD CFRTP skin forming process and establish process conditions suitable for repeated application.
  - Provide forming process support for final prototype fabrication.

- **Phased R&D Implementation Plan and SMART Milestones**

Detailed Task		Institution	Country	Phase 1		Phase 2 1st Year				Phase 2 2nd Year				Phase 2 3rd Year	
				Q1	Q2	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
WP1. Concept Design of Hybrid Ship Chock Liner															
1-1	Investigation of application environment for ship engines and chock liners	3D Factory	KOR												
1-2	Conceptual structural design of hybrid chock liner And 3D printing of the PPS part sample	3D Factory	KOR												
1-3	Definition of UD tape based composite–3D printing joining concept	Fraunhofer ICT	GER												
1-4	Preliminary research on the joining of 3D printed parts with UD tapes using 3DSW	Fraunhofer ICT	GER												
1-5	Preliminary review of PPS UD CFRTP skin forming	UNIST	KOR												
WP2. Composite–3D Printing Joining Process Development															
2-1	Definition of material requirements for chock liners	3D Factory	KOR												
2-2	Analysis of PPS composite 3D printing interfacial characteristics	Fraunhofer ICT	GER												
2-3	Design of joining process concept and process flow	Fraunhofer ICT	GER												
2-4	Design of fixation structure and load transfer path for chock liners	3D Factory	KOR												
2-5	Evaluation of material properties of PPS UD CFRTP skin	UNIST	KOR												
2-6	Conceptual and basic design of joining jigs	3D Factory Fraunhofer ICT	KOR GER												
2-7	Refinement of joining process conditions and verification of applicability	Fraunhofer ICT	GER												
2-8	Verification of joining compatibility between skin and printed structure	UNIST	KOR												
WP3. Fabrication of Hybrid Chock Liner Prototypes															
3-1	Development of PPS UD CFRTP skin forming process	UNIST Fraunhofer ICT	KOR GER												
3-2	Development of large-scale PPS 3D printing process	3D Factory	KOR												
3-3	Manufacturing of molds for skin forming	3D Factory	KOR												
3-4	Manufacturing of composite–3D printing joining jigs	3D Factory Fraunhofer ICT	KOR GER												
3-5	Fabrication of hybrid chock liner prototype	3D Factory	KOR												
3-6	LCA screening for the joining process and ship chock liner	Fraunhofer ICT	GER												
WP4. Fabrication of Final Prototypes and Technology Transfer															
4-1	Optimization of PPS UD CFRTP skin forming process	UNIST	KOR												
4-2	Optimization of composite–3D printing joining process	3D Factory Fraunhofer ICT	KOR GER												
4-3	Fabrication of final hybrid chock liner prototype	3D Factory	KOR												
4-4	Technology transfer and documentation of joining and manufacturing processes	3D Factory Fraunhofer ICT	KOR GER												

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- **Detailed R&D Work Plan and Schedule by Work Package**

Phase 1 (6 month)	**Phase 1. Concept Design of Hybrid Ship Chock Liner**		
	3D Factory	**Fraunhofer ICT**	**UNIST**
	WP 1-1 Application environment analysis	WP 1-3 UD tape composite–3D printing joining concept definition	WP 1-4 PPS UD CFRTP skin forming preliminary review
	WP 1-2 Hybrid chock liner structural design	WP 1-4 Joining of 3D printed parts with UD tapes using 3DSW research on preliminary level	
	WP 1-2 PPS part 3D printing sample		
Phase 2 1st Year (12 month)	**Phase 2-1. Joining Process Development**		
	3D Factory	**Fraunhofer ICT**	**UNIST**
	WP 2-1 Material requirements definition	WP 2-2 PPS composite 3D printing interface analysis	WP 2-5 PPS UD CFRTP skin property evaluation
	WP 2-4 Fixation structure & load transfer design	WP 2-3 Joining process concept & flow design	
	WP 2-6 Joining jig basic design		
Phase 2 2nd Year (12 month)	**Phase 2-2. WP2 (Refinement) + WP3. Prototype Fabrication**		
	3D Factory	**Fraunhofer ICT**	**UNIST**
	WP 3-2 Large-scale PPS 3D printing process	WP 2-7 Joining condition refinement & applicability check	WP 2-8 Skin–printed structure joining compatibility
	WP 3-3 Skin forming mold fabrication	WP 3-6 LCA screening for joining process and ship chock liner	WP 3-1 PPS UD CFRTP skin forming process development
	WP 3-4 Composite–3D printing joining jig fabrication		
Phase 2 3rd Year (6 month)	**Phase 2-3. WP3 (Completion) + WP4. Final Prototype & Transfer**		
	3D Factory	**Fraunhofer ICT**	**UNIST**
	WP 3-5 Hybrid chock liner prototype fabrication	WP 4-2 Composite–3D printing joining optimization	WP 4-1 PPS UD CFRTP skin forming optimization
	WP 4-3 Final hybrid chock liner fabrication	WP 4-4 Technology transfer & process documentation	

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**International Research Institute: Fraunhofer ICT** • Fraunhofer ICT serves as the key international research partner in this international joint R&D project, leading the development of composite–3D printing joining technologies. Based on its long-standing expertise in composite manufacturing and additive manufacturing technologies, Fraunhofer ICT conducts joint research on core joining technologies that are difficult to secure through domestic efforts alone. - Research on composite–3D printing joining mechanisms : Analyze interfacial characteristics between PPS-based composite skins and 3D-printed structures, and establish technical concepts for stable joining. - Development of joining processes and jig design technologies : Design joining process flows and derive concepts and design criteria for joining jigs applicable to chock liner manufacturing. - Support for process applicability and stability verification : Review the applicability of the developed joining processes and propose technical improvement measures from a manufacturing process perspective. - Technical consulting and joint research activities : Provide technical consulting on joining technologies, process design, and manufacturing strategies through joint research with domestic research institutes and the Korean SME lead organization. - Technology transfer and technical documentation : Transfer the developed joining technologies to the Korean SME lead organization in the later stage of the project and provide process manuals and technical documentation. • Through these activities, Fraunhofer ICT plays a critical role as a core technical partner supporting the internalization of advanced joining technologies through international joint research. **Domestic Research Institute: UNIST** • UNIST serves as the domestic research institute responsible for developing PPS UD CFRTP skin forming processes and verifying material properties in this joint R&D project. By supporting product development by the Korean SME lead organization, UNIST conducts technical validation and applicability assessments from both material and process perspectives. - Development of PPS UD CFRTP skin forming processes : Develop skin forming processes based on IR heating and compression molding, and derive process conditions applicable to the contact surface of chock liners. - Evaluation of composite skin properties and process characteristics : Analyze the mechanical and thermal properties of formed PPS UD CFRTP skins and verify their applicability. - Support for integration between forming and joining processes : Review the compatibility between composite skin forming processes and 3D-printed structure joining processes, and provide technical support for process integration. - Process technology support for prototype manufacturing : Support skin forming and process application during prototype fabrication and provide technical feedback to ensure process stability. - Process optimization and reproducibility : Optimize skin forming processes in the later stage of the project and establish process conditions suitable for repeated application. • Through these contributions, UNIST plays a key role as a core domestic research institute responsible for verifying process applicability under domestic conditions and securing experimental data for this project.

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Expected R&R of Korean SME Partner	**Korean SME Lead Organization: 3D Factory** **MOTH Co (Lead Organization)** • 3D Factory serves as the Korean SME lead organization in this international joint R&D project, taking overall responsibility for project execution and leading the industrial application of the developed technologies. Based on its large-scale 3D printing manufacturing capabilities, 3D Factory integrates the technologies developed by the research institutes into practical ship chock liner products. - Overall project management and coordination of joint R&D activities : Oversee collaborative R&D activities among domestic and international partners, manage project schedules, and monitor the achievement of phase-specific objectives. - Product definition and structural design of chock liners : Define performance requirements for chock liners based on marine engine installation conditions and conduct structural design of PPS-based composite–3D printing hybrid chock liners. - Fabrication of large-scale 3D-printed structures and process integration : Manufacture chock liner structures using large-scale 3D printing and integrate the composite skin forming process into the overall manufacturing workflow. - Prototype fabrication and verification of manufacturability and applicability : Fabricate hybrid chock liner prototypes and verify their manufacturability and industrial applicability. - Technology internalization and linkage to commercialization : Internalize core technologies secured through joint R&D and establish commercialization strategies for future mass production and market deployment. • Through these activities, 3D Factory plays a central role as the key lead organization that connects R&D outcomes to actual products and industrial applications within this project.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Outcomes** • Through this project, it is expected that manufacturing technologies for PPS-based composite–3D printing hybrid ship chock liners will be secured. The key expected technical outcomes are as follows: - Securing novel chock liner structures and manufacturing technologies that overcome the limitations of conventional PA block machining and epoxy-based grouting methods - Establishment of hybrid chock liner design and manufacturing processes integrating PPS UD CFRTP skins with large-scale 3D-printed structures - Internalization of core manufacturing technologies through joint research on composite–3D printing joining technologies - Verification of practical applicability through pilot-level chock liner prototype fabrication and achievement of Technology Readiness Level (TRL) 6 • Through these outcomes, the project provides a technological foundation to upgrade chock liners from simple installation components to high-performance, customized marine equipment. **Expected Industrial and Economic Impact** • The outcomes of this project are expected to contribute to strengthening competitiveness across the shipbuilding and marine industries, particularly within the SME-centered marine equipment sector. - Establishment of a foundation for replacing imported or conventional material-based chock liners with domestically developed hybrid chock liners - Reduction of manufacturing lead time and improvement of maintenance efficiency through large-scale 3D printing-based customized manufacturing - Creation of sustained demand centered on the maintenance, repair, and overhaul (MRO) market, considering the relatively short replacement cycle of chock liners

	- Expansion of high value-added marine equipment product portfolios for domestic SMEs **Expected Impact on Global Collaboration** • Through collaboration with overseas research institutes, this project is expected to achieve the following impacts: - Internalization of core joining technologies through joint development, rather than simple technology acquisition - Enhancement of converged technological capabilities in composite materials and additive manufacturing for domestic companies and research institutes - Establishment of a foundation for expanding international joint research into various industrial sectors, including shipbuilding and marine applications **Expected Societal and Strategic Impact** • The outcomes of this project are also expected to generate strategic and societal impacts at the national and industrial levels. - Promotion of digital manufacturing and customized production technologies within the shipbuilding and marine industries - Expansion of high-performance composite material applications beyond conventional metal- and commodity-material-based structures - Strengthening of industrial ecosystem competitiveness through SME-led technology development - Provision of a technological foundation for future expansion into environmentally friendly and high-efficiency marine systems

6. References

(1) BECK, Björn, et al. 3D skeleton winding (3DSW)–Overmolding of wound continuous fiber reinforcements. In: AIP Conference Proceedings. AIP Publishing LLC, 2024. p. 020011.

(2) ITW Performance Polymers, Design and Installation of Epoxy Chocks in Industrial Applications (Technical Guide 643F). ITW Performance Polymers (Chockfast), USA, 2019.

(3) Piaseczny, L, Marine engine seating on polymer–metal chocking. Combustion Engines, Vol. 173, No. 2, pp. 67–72, 2008.

(4) JIANG, Zhenyu, et al. Study on friction and wear behavior of polyphenylene sulfide composites reinforced by short carbon fibers and sub-micro TiO2 particles. *Composites Science and Technology*, 2008, 68.3-4: 734-742.

(5) SIDDIQUE, Amna, et al. A review of joining techniques for thermoplastic composite materials. *Journal of Thermoplastic Composite Materials*, 2023, 36.8: 3417-3454.

(6) XU, Zhiwu, et al. Insights into the influence of welding energy on the ultrasonic welding of glass fibre-reinforced PPS composites. *Composites Science and Technology*, 2024, 256: 110767.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization